

Step 1: Assign ip for Pc and Ping another Network Pc's

```
C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=3ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=3ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=2ms TTL=125

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Documents and Settings\user>_
```

```
C:\Documents and Settings\user>ncpa.cpl

C:\Documents and Settings\user>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=1ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=1ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time<1ms TTL=128
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>
```

```

C:\Documents and Settings\user>ncpa.cpl

C:\Documents and Settings\user>PING 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=4ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=3ms TTL=124

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>PING 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time<1ms TTL=128
Reply from 172.36.0.10: bytes=32 time<1ms TTL=128
Reply from 172.36.0.10: bytes=32 time<1ms TTL=128
Reply from 172.36.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>

```

Router 1:

```

Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/8 is directly connected, Ethernet0/1
L       12.0.0.1/32 is directly connected, Ethernet0/1
    23.0.0.0/24 is subnetted, 1 subnets
S       23.0.0.0 [1/0] via 12.0.0.2
    24.0.0.0/24 is subnetted, 1 subnets
S       24.0.0.0 [1/0] via 12.0.0.2
    26.0.0.0/24 is subnetted, 1 subnets
S       26.0.0.0 [1/0] via 12.0.0.2
    32.0.0.0/24 is subnetted, 1 subnets
S       32.0.0.0 [1/0] via 12.0.0.2
    45.0.0.0/24 is subnetted, 1 subnets
S       45.0.0.0 [1/0] via 12.0.0.2
    172.36.0.0/24 is subnetted, 1 subnets
S       172.36.0.0 [1/0] via 12.0.0.2
    192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.50.20.0/24 is directly connected, Ethernet0/0
L       192.50.20.1/32 is directly connected, Ethernet0/0
S       223.0.50.0/24 [1/0] via 12.0.0.2
R1#

```

Router 2:

```
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#interface e0/1
R2(config-if)#ip address 26.0.0.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#ip route 192.50.20.0 255.255.255.0 12.0.0.1
R2(config)#ip route 32.0.0.0 255.255.255.0 23.0.0.2
R2(config)#ip route 45.0.0.0 255.255.255.0 24.0.0.2
R2(config)#ip route 172.36.0.0 255.255.255.0 23.0.0.2
R2(config)#ip route 223.0.50.0 255.255.255.0 26.0.0.2
R2(config)#
R2(config)#end
R2#wr
*Jul 21 16:46:29.861: %SYS-5-CONFIG_I: Configured from console by console
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 172.36.0.0 255.255.255.0 24.0.0.2
R2(config)#
*Jul 21 16:50:28.017: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config)#
```

Router 3:

```
R3(config-if)#ip address 23.0.0.2 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#interface e0/1
R3(config-if)#ip address 32.0.0.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#ip route 192.50.20.0 255.255.255.0 23.0.0.1
R3(config)#ip route 12.0.0.0 255.255.255.0 23.0.0.1
R3(config)#ip route 24.0.0.0 255.255.255.0 23.0.0.1
R3(config)#ip route 26.0.0.0 255.255.255.0 23.0.0.1
R3(config)#ip route 45.0.0.0 255.255.255.0 32.0.0.2
R3(config)#ip route 172.36.0.0 255.255.255.0 32.0.0.2
R3(config)#ip route 223.0.50.0 255.255.255.0 23.0.0.1
R3(config)#
R3(config)#end
R3#wr
*Jul 21 16:50:50.394: %SYS-5-CONFIG_I: Configured from console by console
R3#wr
*Jul 21 16:51:11.537: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R3#wr
```

Router 4:

```
R4(config-if)#ip address 24.0.0.2 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
R4(config)#interface e0/1
R4(config-if)#ip address 45.0.0.1 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
R4(config)#ip route 192.50.20.0 255.255.255.0 24.0.0.1
R4(config)#ip route 12.0.0.0 255.255.255.0 24.0.0.1
R4(config)#ip route 23.0.0.0 255.255.255.0 24.0.0.1
R4(config)#ip route 26.0.0.0 255.255.255.0 24.0.0.1
R4(config)#ip route 32.0.0.0 255.255.255.0 45.0.0.2
R4(config)#ip route 172.36.0.0 255.255.255.0 45.0.0.2
R4(config)#ip route 223.0.50.0 255.255.255.0 24.0.0.1
R4(config)#
R4(config)#end
R4#wr
*Jul 21 16:51:40.180: %SYS-5-CONFIG_I: Configured from console by console
R4#wr
*Jul 21 16:52:23.175: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R4#wr
```

Router 5:

```
R5(config)#interface e0/1
R5(config-if)#ip address 32.0.0.2 255.255.255.0
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
R5(config)#interface e0/2
R5(config-if)#ip address 45.0.0.2 255.255.255.0
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
R5(config)#ip route 192.50.20.0 255.255.255.0 32.0.0.1
R5(config)#ip route 12.0.0.0 255.255.255.0 32.0.0.1
R5(config)#ip route 23.0.0.0 255.255.255.0 32.0.0.1
R5(config)#ip route 24.0.0.0 255.255.255.0 45.0.0.1
R5(config)#ip route 26.0.0.0 255.255.255.0 45.0.0.1
R5(config)#ip route 223.0.50.0 255.255.255.0 45.0.0.1
R5(config)#
R5(config)#end
R5#wr
*Jul 21 16:52:13.123: %SYS-5-CONFIG_I: Configured from console by console
R5#wr
*Jul 21 16:52:44.295: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R5#wr
```

Router 6:

```
R6(config-if)#ip address 223.0.50.1 255.255.255.0
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#
R6(config)#interface e0/1
R6(config-if)#ip address 26.0.0.2 255.255.255.0
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#
R6(config)#ip route 192.50.20.0 255.255.255.0 26.0.0.1
R6(config)#ip route 12.0.0.0 255.255.255.0 26.0.0.1
R6(config)#ip route 23.0.0.0 255.255.255.0 26.0.0.1
R6(config)#ip route 24.0.0.0 255.255.255.0 26.0.0.1
R6(config)#ip route 32.0.0.0 255.255.255.0 26.0.0.1
R6(config)#ip route 45.0.0.0 255.255.255.0 26.0.0.1
R6(config)#ip route 172.36.0.0 255.255.255.0 26.0.0.1
R6(config)#
R6(config)#end
R6#wr
*Jul 21 16:52:36.272: %SYS-5-CONFIG_I: Configured from console by console
R6#wr
*Jul 21 16:57:12.840: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R6#wr
```